H

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
Higher 2

CANDIDATE NAME				
CLASS		TUTOR'S NAME		
CENTRE NUMBER	S		INDEX NUMBER	

PHYSICS 9478/02

Paper 2 Structured Questions **16 September 2026**

2 hours

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name, class, Centre number and index number in the spaces at the top of this page.
Write in dark blue or black pen on both sides of the paper.
You may use a HB pencil for any diagrams, graphs.
Do not use staples, paper clips, glue or correction fluid.

The use of an approved scientific calculator is expected, where appropriate.
Answer **all** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	/ 10
2	/ 8
3	/ 10
4	/ 14
5	/ 11
6	/ 22
Total	/ 75

This document consists of **19** printed pages.

Data

speed of light in free space	c	=	3.00×10^{8} m s^{-1}
permeability of free space	μ_0	=	$4\pi \times 10^{-7}$ H m^{-1}
permittivity of free space	ε_0	=	8.85×10^{-12} F m^{-1}
	$(\frac{1}{4\pi\varepsilon_0}$	=	8.99×10^{9} m F^{-1})
elementary charge	e	=	1.60×10^{-19} C
the Planck constant	h	=	6.63×10^{-34} J s
unified atomic mass constant	u	=	1.66×10^{-27} kg
rest mass of electron	m_e	=	9.11×10^{-31} kg
rest mass of proton	m_p	=	1.67×10^{-27} kg
molar gas constant	R	=	8.31 J K^{-1} mol^{-1}
Avogadro constant	N_A	=	6.02×10^{23} mol^{-1}
Boltzmann constant	k	=	1.38×10^{-23} J K^{-1}
gravitational constant	G	=	6.67×10^{-11} N m^2 kg^{-2}
acceleration of free fall	g	=	9.81 m s^{-2}

Formulae

uniformly accelerated motion	s	=	$ut + \frac{1}{2}at^2$
	v^2	=	$u^2 + 2as$
work done on / by gas	W	=	$p\,\Delta V$
pressure	p	=	$\frac{F}{A}$
gravitational potential	ϕ	=	$-\frac{GM}{r}$
temperature	T / K	=	$T / °C + 273.15$
pressure of an ideal gas	p	=	$\frac{1}{3}\frac{Nm}{V} < c^2 >$
mean translational kinetic energy of an ideal gas particle	E	=	$\frac{3}{2}kT$
displacement of particle in s.h.m.	x	=	$x_0 \sin \omega t$
velocity of particle in s.h.m.	v	=	$v_0 \cos \omega t = \pm\omega\sqrt{\left(x_0{}^2 - x^2\right)}$
electric current	I	=	$Anvq$
resistors in series	R	=	$R_1 + R_2 + \ldots$
resistors in parallel	$1/R$	=	$1/R_1 + 1/R_2 + \ldots$
capacitors in series	$1/C$	=	$1/C_1 + 1/C_2 + \ldots$
capacitors in parallel	C	=	$C_1 + C_2 + \ldots$
energy in a capacitor	U	=	$\frac{1}{2}QV = \frac{1}{2}\frac{Q^2}{C} = \frac{1}{2}CV^2$
charging a capacitor	Q	=	$Q_0\left[1 - e^{-\frac{t}{\tau}}\right]$
discharging a capacitor	Q	=	$Q_0 e^{-\frac{t}{\tau}}$
RC time constant	τ	=	RC
electric potential	V	=	$\frac{Q}{4\pi\varepsilon_0 r}$
alternating current / voltage	x	=	$x_0 \sin \omega t$
magnetic flux density due to a long straight wire	B	=	$\frac{\mu_0 I}{2\pi d}$
magnetic flux density due to a flat circular coil	B	=	$\frac{\mu_0 NI}{2r}$
magnetic flux density due to a long solenoid	B	=	$\mu_0 nI$
energy states for quantum particle in a box	E_n	=	$\frac{h^2}{8mL^2}n^2$
radioactive decay	x	=	$x_0 e^{-\lambda t}$
radioactive decay constant	λ	=	$\frac{\ln 2}{t_{\frac{1}{2}}}$

Answer **all** the questions in the spaces provided.

1 **(a)** State the conditions required for a body to be in equilibrium.

.....

.....

.....

.....[2]

(b) Block A of mass 1.7 kg is connected by a light inextensible string to a counterweight B of mass *M* via a system of two massless and frictionless pulleys P and Q. Block A is balanced on a smooth plane inclined at 45°.

Pulley Q is a double pulley made up of two fixed wheels of different radii attached to a common axle, with the radii of the smaller and larger wheels *r* and 3*r* respectively as shown in Fig. 1.1.

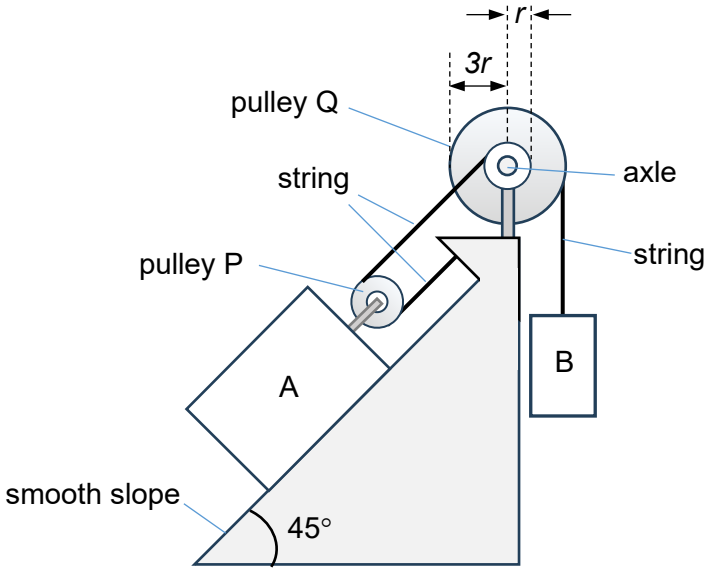


Fig.1.1

(i) Block A is in equilibrium. In Fig. 1.2, draw and label all forces acting on block A.

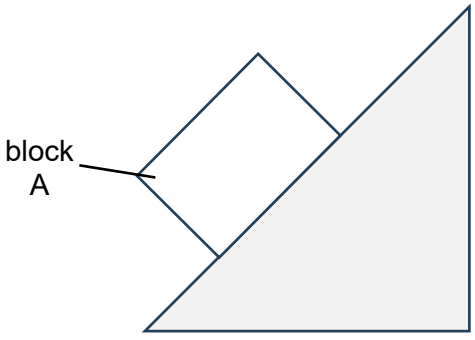


Fig.1.2

[2]

(ii) Calculate the tension in the string connected to pulley P.

tension = ... N [2]

(iii) Determine the mass *M* of counterweight B.

M = ... kg [2]

(iv) Suggest and explain a change that can be made to the pulley such that block A will accelerate up the slope.

.....

.....

.....

.....[2]

[Total: 10]

2 A plane is flying with a velocity of 240 m s^{-1} at an angle of 30° with respect to the horizontal, as shown in Fig. 2.1. At an altitude of 2.0 km, a projectile is released from the plane. The projectile hits the target on the ground. Assume air resistance is negligible.

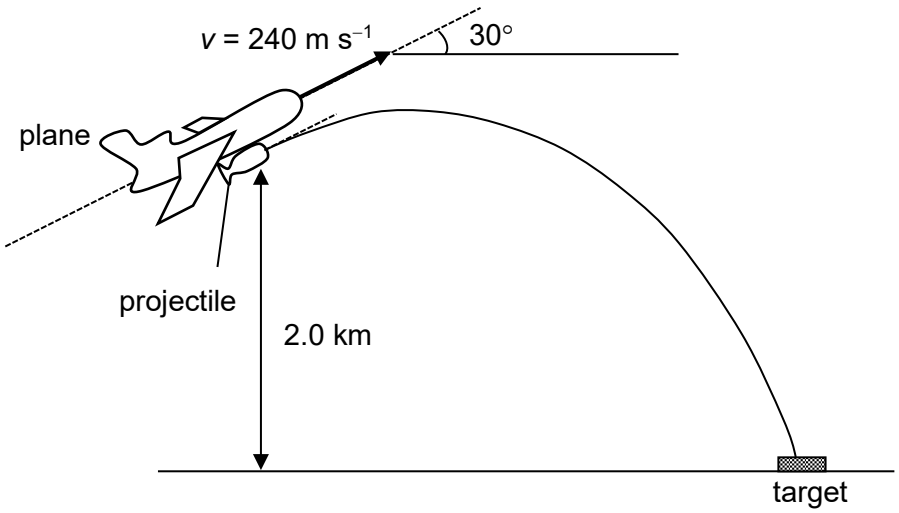


Fig. 2.1

(a) Determine the horizontal displacement of the projectile from the point of release to the target.

horizontal displacement = ... m [3]

(b) Determine the speed of the projectile just before it hits the target.

speed = ... m s^{-1} [2]

(c) (i) Without further calculation, on Fig. 2.2, sketch the variation of vertical velocity v_y with time t of the projectile from the time it was released to the time it hits the target. Label this line **P**. Include values on your axes. [1]

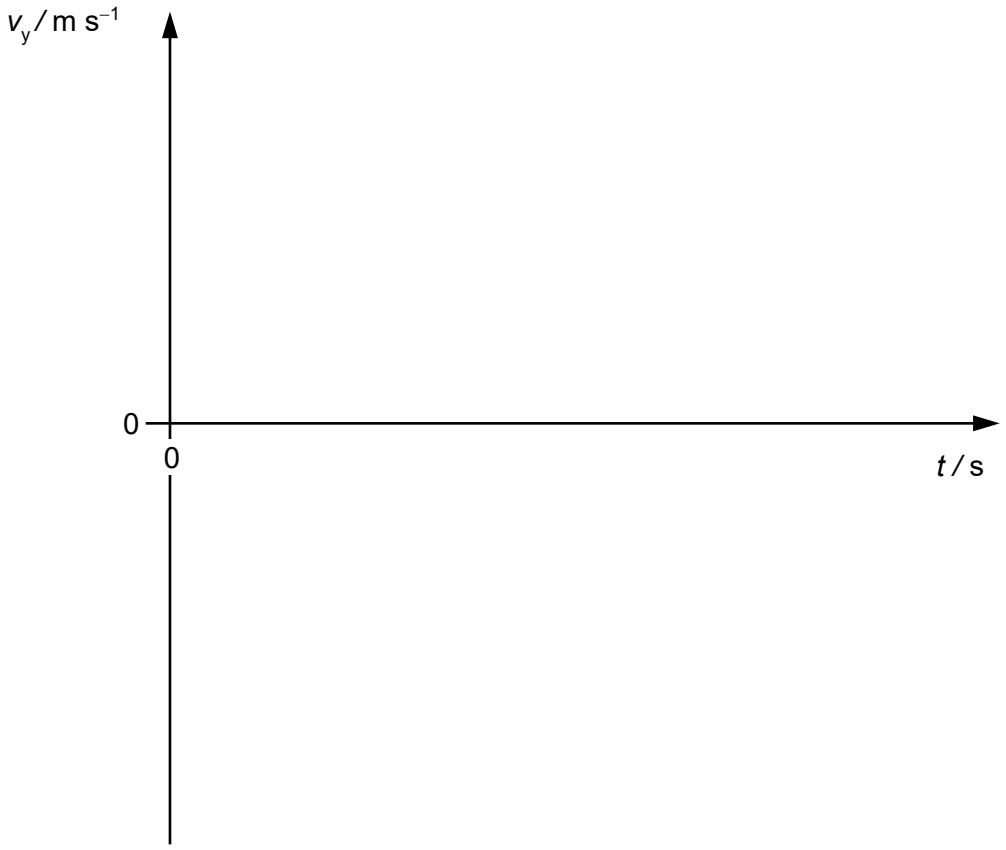


Fig. 2.2

(ii) On Fig. 2.2, sketch the variation of vertical velocity v_y with time t of the projectile if air resistance was **not** negligible. Label this line **Q**. No calculations are required. [2]

[Total: 8]

3 **(a)** PQRS comprises of a rough slope PQR of height 1.1 m, and smooth horizontal surface RS. PQR of total surface length 2.7 m is made up of a straight section PQ and curved section QR as shown in Fig. 3.1.

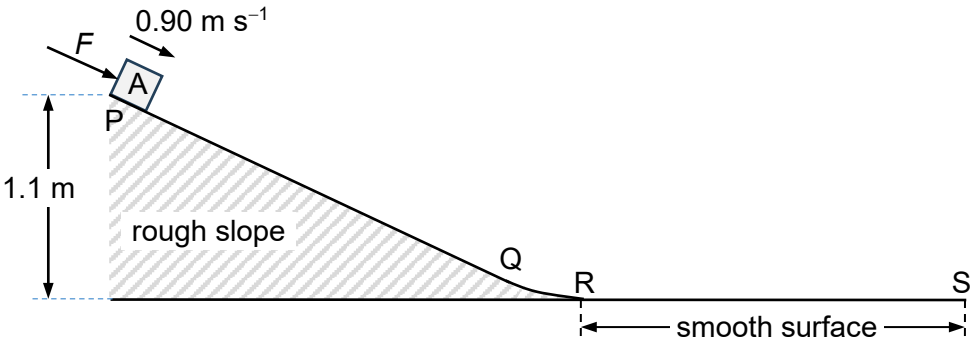


Fig. 3.1

Block A of mass 2.0 kg has an initial speed of 0.90 m s^{-1} at point P on rough slope. A force *F* is applied to block A as it moves along PQR. The magnitude of the *F* is 11 N and its direction is the same as the object's velocity at any instant along surface PQR.

The frictional force between the block A and the surface PQR may be taken to be constant at 6.4 N.

(i) Determine the speed of the block when it reaches R.

speed = ... m s^{-1} [3]

(ii) Use Newton's second law to explain how the speed of block A varies along PQRS.

.....

.....

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.....

.....[3]

(b) Block X of mass 0.27 kg moves with a speed 0.67 m s^{-1} on a smooth horizontal surface. Block Y of mass 0.34 kg moves with speed 0.46 m s^{-1} in the opposite direction. A light spring of natural length 13.0 cm and force constant 120 N m^{-1} is attached to block Y as shown in Fig. 3.2.

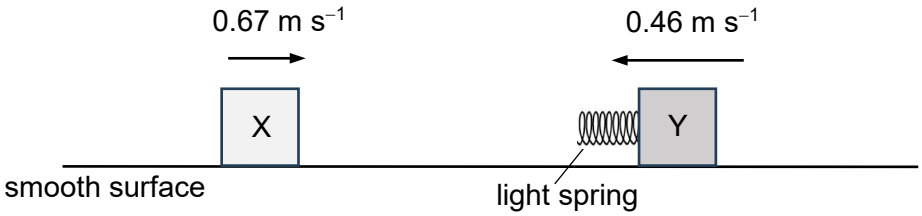


Fig. 3.2

When block X collides elastically head-on with block Y, there will be an instant when the spring experiences maximum compression.

Determine the closest distance of approach of the two blocks.

closest distance of approach = ... m [4]

[Total: 10]

4 **(a)** **(i)** State **one** difference and **one** similarity between the oscillations of a stationary sound wave and a progressive sound wave.

Difference: .....

.....

.....

Similarity: .....

.....

.....

[2]

(ii) Fig. 4.1 represents the displacement-time graph of two progressive sound waves travelling together through the same space.

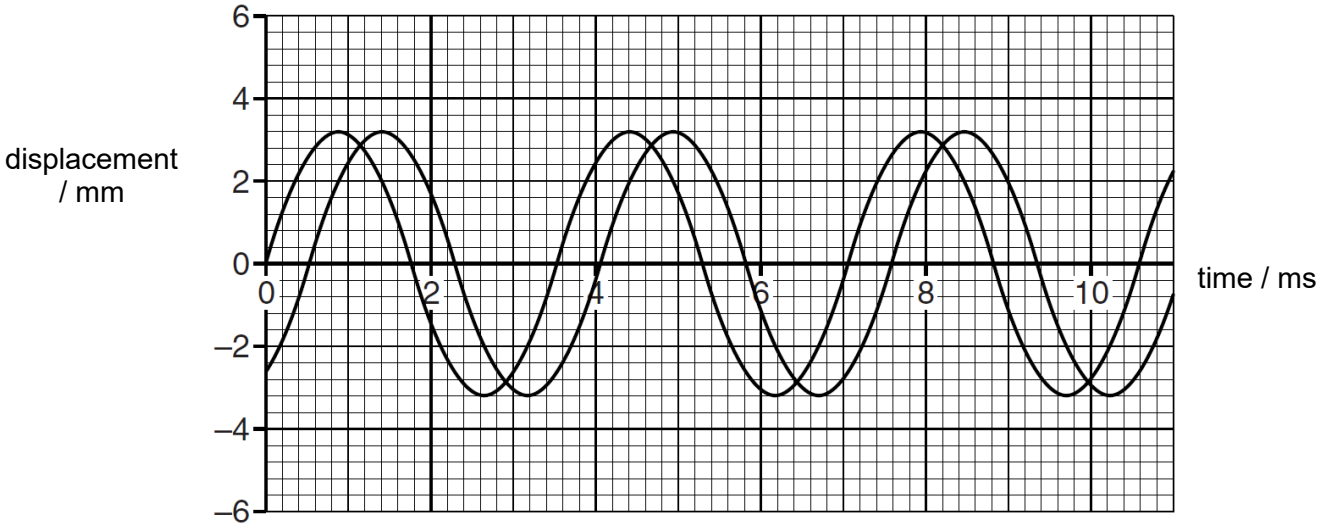


Fig. 4.1

1. State and explain whether Fig. 4.1 shows that the waves are coherent.

.....

.....

.....[1]

2. Use Fig. 4.1 to determine the phase difference between the waves.

phase difference = ... rad [1]

(iii) A sound wave travels along the axis of a tube that is open at both ends, as shown in Fig. 4.2. A stationary sound wave is formed in the tube.

The lowest frequency possible for the stationary sound wave in this tube is *f*.

On Fig. 4.2, mark all the positions of the displacement nodes (use the letter N) and the displacement antinodes (use the letter A) of a stationary sound wave for a frequency of 2*f*.

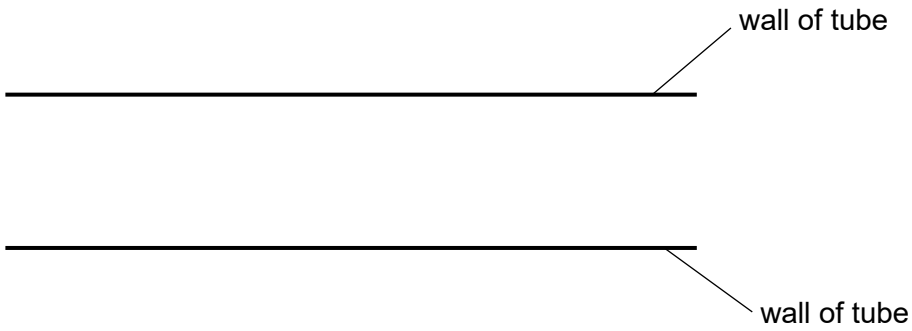


Fig. 4.2

[1]

(iv) With reference to the movement of air molecules on either side of a displacement node, explain why there is maximum variation in pressure at the nodes.

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.....

.....

.....[2]

(b) Fig. 4.3 shows the arrangement for an experiment involving superposition of microwaves.

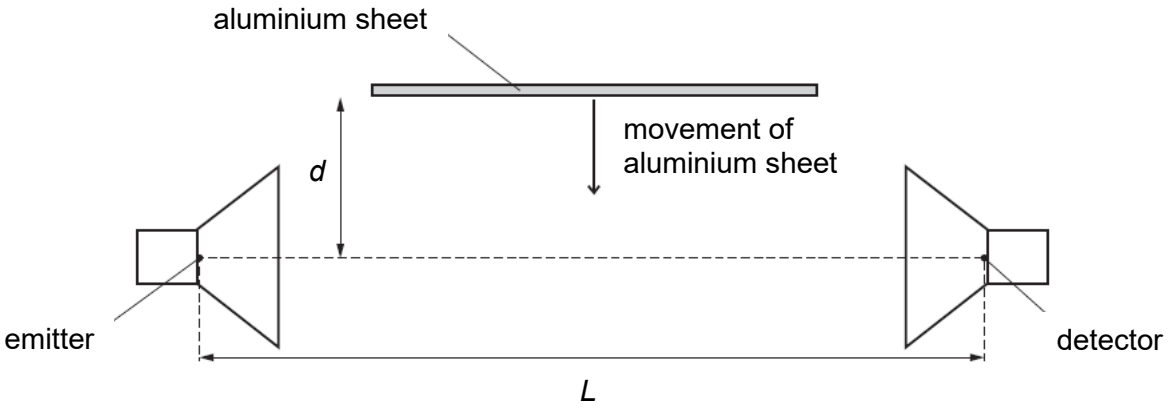


Fig. 4.3 (not to scale)

A microwave emitter and detector face each other with a distance *L* apart. A reflecting aluminium sheet is placed at a vertical distance *d* above the direct path between the emitter and the detector.

(i) When the aluminium sheet is moved in the direction shown by the arrow, the intensity of microwaves received by the detector fluctuates.

Describe and explain this variation in the microwave intensity.

.....

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.....[2]

(ii) As d decreases, the microwave detector has a maximum reading at d = 21.2 cm and the next maximum at d = 17.7 cm. The distance L stays constant at 80.0 cm.

Determine the wavelength λ of the microwaves.

λ = ... m [3]

(iii) Microwaves undergo a phase change of π rad when they reflect.

When d is much less than the wavelength of the microwaves, it is observed that the detector reading is a minimum.

Suggest why this observation demonstrates that microwaves must undergo a phase change on reflection from the aluminium sheet.

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.....[2]

[Total: 14]

5 **(a)** State what is meant by the mass defect of a nucleus.

.....

.....

.....[2]

(b) A stationary nucleus of uranium-238 ($^{238}_{92}U$) undergoes alpha decay to produce an excited nucleus of thorium-234 ($^{234}_{90}Th^*$). The kinetic energy of the emitted alpha particle is 4.151 MeV. A gamma-ray photon is then immediately emitted from the excited thorium nucleus after the alpha decay.

Assume that the rebound kinetic energy of the thorium nucleus is negligible.

Fig. 5.1 shows the masses of some of the nuclides involved in the decay reaction.

nuclide	nuclide mass/ u
$^{4}_{2}\alpha$	4.000407
$^{234}_{90}Th$	233.915174

Fig. 5.1

The total energy released in the decay of the nucleus of uranium-238 is 4.274 MeV.

(i) Calculate the mass, in u, of the uranium-238 nuclide.

mass = ... u [3]

(ii) Determine a value for the wavelength of the gamma radiation emitted during the decay of the uranium-238 nucleus.

wavelength = ... m [3]

(iii) In practice, the rebound kinetic energy of the thorium nucleus is **not** negligible.

Explain, without further calculation, how your answer in **(b)(ii)** compares with the true wavelength of gamma radiation emitted during the decay of the uranium-238 nucleus.

.....

.....[1]

(c) Gamma radiation emitted during the decay of a sample of uranium-238 has a single wavelength.

Nuclei of cobalt-60 ($^{60}_{27}\mathrm{Co}$) decay by beta emission and also emit gamma radiation in the process.

Suggest why there is **not** a single wavelength for the gamma radiation emitted during the decay of a sample of cobalt-60.

.....

.....

.....

.....

.....[2]

[Total: 11]

6 Read the passage and answer the questions that follow.

The Juno spacecraft is launched by NASA in 2011, and its mission is to study Jupiter's origin, interior structure, atmosphere, and magnetosphere to understand the solar system's formation by measuring composition, gravity, and magnetic fields with its myriad of sensors on board.

This is also NASA's first mission to Jupiter to use solar panels rather than radioisotope thermoelectric generators (RTGs) for powering the spacecraft. Jupiter orbits the Sun at a much greater distance than Earth, making solar power generation a significant engineering challenge.

Solar radiation spreads out as it travels from the Sun. The solar intensity I at a distance d from the Sun follows the inverse square law. The solar intensity on Jupiter is only 3.72 % that on Earth. To compensate for the low intensity at Jupiter, Juno is equipped with three large solar array wings, as shown in Fig 6.1.

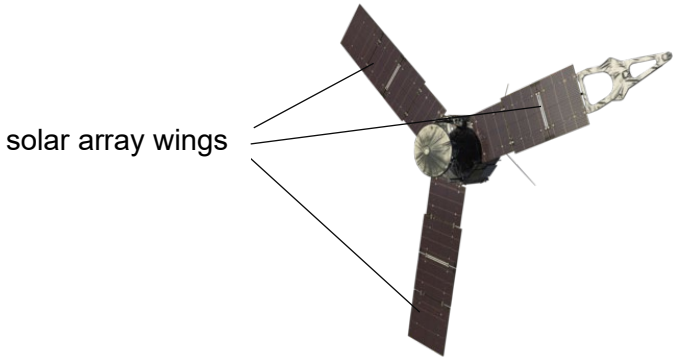


Fig. 6.1

The electrical power P generated by the solar panels on the wings is given by the equation:

$$P = I A \eta$$

where A is the total surface area of the solar panels and η is the efficiency of the solar cells. The efficiency of the cells is affected by the extremely low temperatures in the outer solar system.

Jupiter has an intense radiation belt above its equator that spans a distance of $2R_J$ to $10R_J$ from its centre, where R_J is the radius of Jupiter. Exposure to excessive radiation can cause damage to the spacecraft and its sensors. To minimize such exposure, Juno follows a highly elliptical polar (pole-to-pole) orbit, as shown in Fig 6.2.

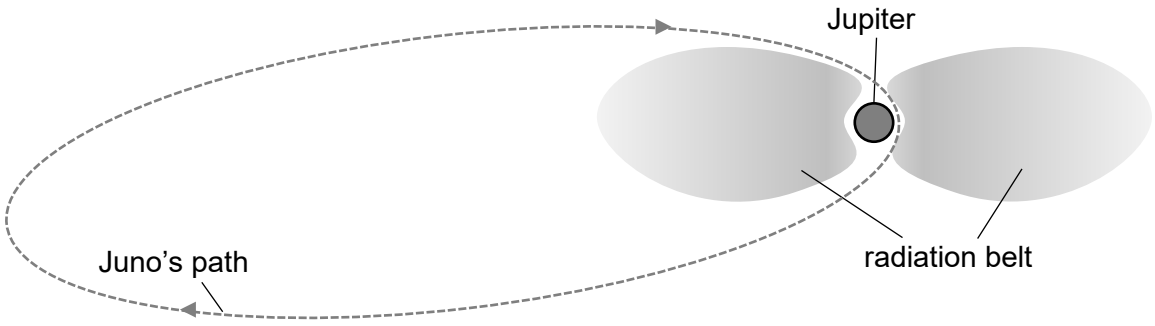


Fig. 6.2

Once Juno enters Jupiter's orbit, its propulsion system is turned off. It is subsequently used sparingly only to correct its path when necessary.

The distance of Juno from the centre of Jupiter in the first three months of 2017 is shown in Fig. 6.3.

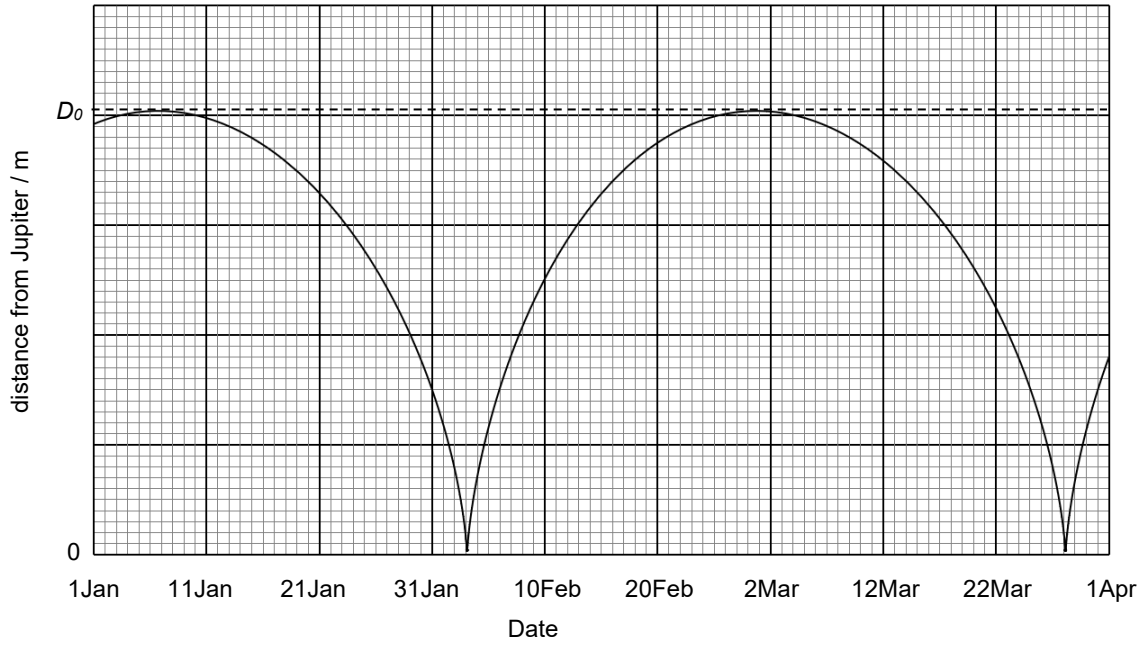


Fig. 6.3

Due to the vast distances involved, it took almost 5 years for Juno to enter Jupiter's orbit from Earth. Communication between Juno and Earth which is conducted via radio waves also experiences significant "light time" delay for signals.

Additional data for Earth, Jupiter and the Juno mission are provided in Table 6.4.

Property	**Value**
Mass of Jupiter M_J	1.90×10^{27} kg
Radius of Jupiter R_J	7.15×10^{7} m
Mean distance* of Earth from the Sun d_E	1.50×10^{11} m
Solar intensity at Earth's distance I_E	1360 W m^{-2}
Total surface area of Juno's solar panels A	60 m^2
Efficiency of solar cells at Jupiter η	24%
Mass of Juno spacecraft m	3625 kg
Perijove (closest distance* to Jupiter) r_p	1.06 R_J
Apojove (furthest distance* from Jupiter) r_a	113 R_J

*center to center

Table 6.4

(a) Explain why the solar intensity I follows an inverse square law as the distance d from the Sun increases.

.....

.....

.....[2]

(b) Using the relation in **(a)** and the data provided, show that the mean distance of Jupiter from the Sun is 7.78×10^{11} m.

[2]

(c) Calculate the total electrical power output generated by Juno's solar panels when Jupiter is at its mean distance from the Sun.

electrical power = ... W [1]

(d) Suggest **one** reason, other than distance from the Sun, why the electrical power output might fluctuate during the mission.

.....

.....[1]

(e) Determine from Fig. 6.3 the period of Juno's elliptical orbit.

period = ... days [1]

(f) Determine the value of D_0 in Fig. 6.3.

D_0 = ... m [1]

(g) Explain why the sum of Juno's kinetic energy and gravitational potential energy remains constant when it is in orbit.

.....

.....

.....[1]

(h) Calculate the difference between the maximum and minimum values for the kinetic energy of Juno during an orbit.

difference = ... J [3]

(i) On Fig. 6.2, indicate point X at which Juno is moving with maximum speed and point Y at which it is moving with minimum speed. [1]

(j) Explain using Fig. 6.2 and your answer in **(i)** how the highly elliptical orbit minimizes the spacecraft's exposure to Jupiter's intense radiation belt.

.....

.....

.....

.....[2]

(k) For a body in a circular orbit, gravitational field strength g is always equal to $\frac{v^2}{r}$, where v is its linear speed and r is the radius of the orbit. In Juno's case, assume g is equal to $\frac{v^2}{r}$ at its perijove and its apojove. Use this relationship to calculate Juno's minimum speed.

minimum speed = ... m s^{-1} [3]

(l) Explain why the relationship $g = \frac{v^2}{r}$ does not hold in general for elliptical orbits.

.....

.....

.....[2]

(m) Using the mean distances of Jupiter and of Earth from the Sun, estimate a minimum value for the "light time" delay for signals to travel from Jupiter to Earth.

delay = ... s [2]

[Total: 22]

End of Paper